

Pairs Trading Strategy Framework with Risk Controls

MA8.402: Mathematics for Finance

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Outline

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- 3 Signal Generation & Risk Management
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- 5 Experimental Results & Visual Analysis
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Introduction to Pairs Trading

- **Statistical Arbitrage:** Exploiting short-term market microstructure noise, liquidity shocks, or transient behavioral biases.
- **Pairs Trading Premise:** Two assets with similar fundamental drivers will maintain a stable long-term relationship.
- **Mechanism:** Short the outperforming asset and buy the underperforming one when prices diverge, expecting mean reversion.
- **Our Core Hypothesis:** Incorporating **Granger causality** into the selection pipeline significantly enhances out-of-sample robustness and risk-adjusted returns over traditional Pearson correlation.

Stochastic Processes & Cointegration

- **Asset Prices:** Modeled as Geometric Brownian Motion (GBM). Individual log-prices $p_{i,t}$ are non-stationary $I(1)$ processes.
- **Cointegration:** Two $I(1)$ series are cointegrated if a linear combination results in a stationary $I(0)$ series.

$$S_t = p_{y,t} - \beta p_{x,t}$$

- **Engle-Granger Two-Step Method:**
 - 1 Estimate cointegrating regression using OLS: $p_{y,t} = \alpha + \beta p_{x,t} + \epsilon_t$
 - 2 Test residuals for stationarity using Augmented Dickey-Fuller (ADF) test.

Modeling the Spread: Ornstein-Uhlenbeck Process

- A stationary spread S_t can be mathematically modeled using the continuous-time **Ornstein-Uhlenbeck (OU)** process:

$$dS_t = \theta(\mu - S_t)dt + \sigma dW_t$$

- μ : long-term equilibrium mean
 - $\theta > 0$: speed of mean reversion
 - σ : volatility of the spread
- **Half-life of Mean Reversion:** Expected time for the spread to halve its deviation from the mean:

$$\tau = \frac{\ln(2)}{\theta}$$

Causality vs. Correlation

- **Correlation Fallacy:** Pearson correlation simply measures the degree to which two variables move together. Correlation does not imply cointegration or causation (spurious regressions).
- **Granger Causality:** Time series X Granger-causes Y if past values of X help predict Y beyond past values of Y alone.
- We use **Vector Autoregression (VAR)** and require strong bi-directional or uni-directional Granger causality ($p\text{-value} < 0.05$).
- **Result:** Filters out spurious correlations driven by exogenous market factors.

- **Dynamic Z-Score:** Based on rolling mean ($\mu_{S,t}$) and volatility ($\sigma_{S,t}$).

$$Z_t = \frac{S_t - \mu_{S,t}}{\sigma_{S,t}}$$

Trade entry when $|Z_t| \geq Z_{\text{entry}}$, exit when $|Z_t| \leq Z_{\text{exit}}$.

- **Inverse Volatility Sizing:** Maintains a constant gross exposure risk profile. Size is scaled inversely to the recent volatility of spread changes.

$$k_t = \frac{E_{\text{target}}}{\max(\sigma_{\Delta S,t}, \sigma_{\min})}$$

Strict capital preservation measures are enforced via overlays:

- **Stop Loss:** Active positions liquidated instantly if cumulative PnL drops below a predefined fraction of allocated capital (e.g., -5%).
- **Time Stop:** Given theoretical half-life τ , positions held longer than $T_{\max}$ (e.g., 30 days) are flattened (mean-reverting relationship breakdown).
- **Value at Risk (VaR):** Rolling 95% historical VaR is monitored to ensure the portfolio's tail risk remains within institutional bounds.

System Architecture Overview

Our modular, object-oriented framework consists of 5 decoupled engines:

- ➊ DataDownloader: yfinance API ingestion with joblib local filesystem caching for rapid backtesting.
- ➋ PairIdentificationEngine: $\mathcal{O}(N^2)$ pairwise correlation matrix, Engle-Granger coint tests, and VAR Granger causality tests.
- ➌ PairSignalEngine: Calculates rolling Z-scores and generates normalized dollar weights via inverse-volatility scaling, capping absolute exposure.
- ➍ PairRiskEngine: Intercepts raw signals to enforce Stop Loss, Time Stop, and VaR overlays before execution.
- ➎ PairBacktestEngine: The accounting ledger. Simulates daily mark-to-market, exact transaction costs (turnover), and computes net PnL/Sharpe.

Experimental Setup & Initial Results

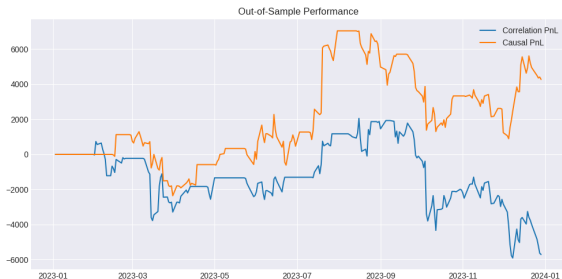
- **Universe:** 20 liquid U.S. large-cap equities.
- **In-Sample:** Jan 2020 – Dec 2022.
- **Out-of-Sample:** Jan 2023 – Jan 2024.

Table: Out-of-Sample Performance Summary

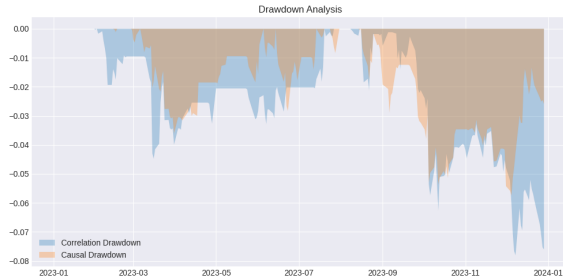
Approach	Total Net PnL (\$)	Max Drawdown	Sharpe Ratio
Correlation	-5,706.35	-7.80%	-0.738
Causal	4,275.69	-5.74%	0.516

The causal approach achieved a positive Sharpe ratio, whereas correlation pairs suffered a severe out-of-sample breakdown.

Performance & Drawdown Analysis

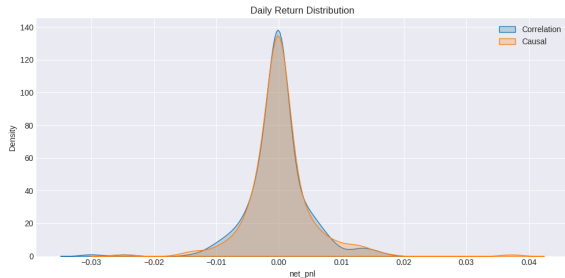


Insight 1: Causal PnL climbs steadily. Correlation PnL drops massively halfway through 2023 due to exogenous shocks.

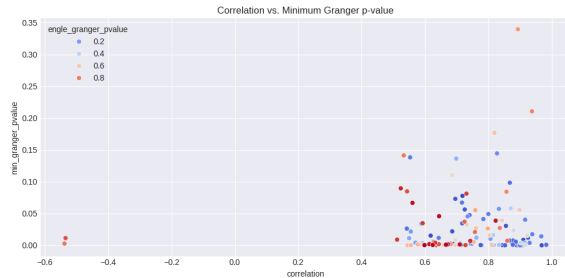


Insight 2: Correlation spends most of OOS in deep drawdown. Causal experiences milder, short-lived drawdowns.

Return Distribution & Causality Verification



Insight 3: Causal returns have a tighter distribution shifted right (consistent gains). Correlation is left-skewed (frequent large losses).

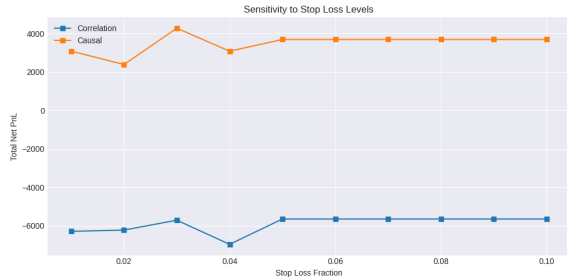


Insight 7: High correlation does NOT guarantee low Granger p-value. Superior pairs exist in the bottom-right quadrant.

Sensitivity: Transaction Costs & Stop Loss

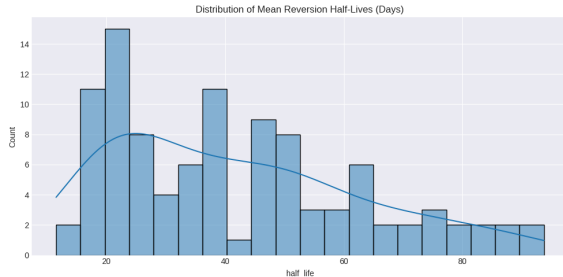


Insight 4: Causal approach remains profitable up to ~ 30 bps/trade. Correlation is unviable across the entire cost spectrum.

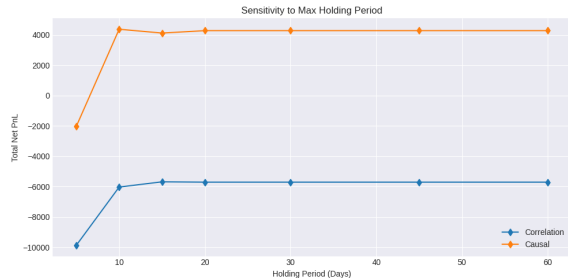


Insight 5: Optimal stop-loss is $\sim 5\%$. Tighter stops ($< 3\%$) choke mean-reversion. Wider stops for correlation just increase terminal losses.

Holding Period Constraints & Half-Life

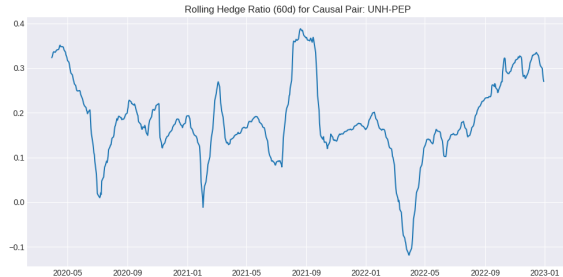


Insight 6: Most cointegrated pairs have a half-life of 10-40 days. This mathematically justifies a 30-day max holding constraint.

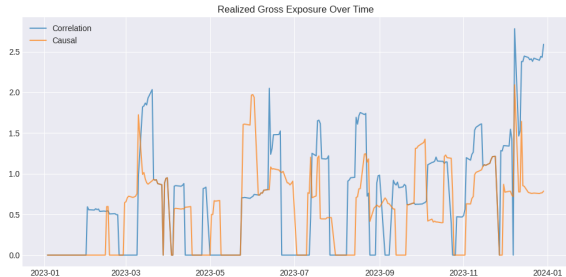


Insight 9: Returns maximize around 30 days. Shorter limits cut mean-reversion early; longer limits introduce uncompensated drift risk.

Stability and Risk Invariants



Insight 8: 60-day rolling Hedge Ratio (β) for causal pairs exhibits bounded, mean-reverting stationarity without structural breaks.



Insight 10: Realized gross exposure strictly obeys the 2.0 maximum ceiling, proving the algorithm's capital bound invariants.

- **Maximum Likelihood Estimation (MLE):** Derive exact discrete-time OU parameters for precise analytical half-life calculation and dynamic thresholding.
- **Kelly Criterion Sizing:** Implement Fractional Kelly position sizing ($f^* = \frac{\mu_{\text{excess}}}{\sigma^2_{\text{portfolio}}}$) to mathematically maximize the compound growth rate.
- **Non-linear Transaction Costs:** Incorporate the Almgren-Chriss market impact model (slippage proportional to $\sqrt{V/\text{ADV}}$) for scaling AUM capacity.

- Simplistic correlation screens routinely fail out-of-sample due to their vulnerability to spurious, historically coincident trends.
- Integrating rigorous causal inference (**Granger causality**) acts as a powerful structural filter, isolating pairs that possess a genuine information-driven link.
- Our **Causal Pairs** framework successfully maintained cointegration out-of-sample, preserving capital through drawdowns and generating a positive Sharpe Ratio (0.516) under strict institutional-grade risk limits.

Thank You!

Questions?